

MATHEMATICS SUBJECTIVE TEST
Class X (Session 2026 – 27)

ANSWER KEY

SECTION-A

1. By using Identity : $(a + b)^2 + (a - b)^2 = 2a^2 + 2b^2$
Here $a = \sqrt{5}$ and $b = \sqrt{3}$, so : $(\sqrt{5} + \sqrt{3})^2 + (\sqrt{5} - \sqrt{3})^2 = 2(\sqrt{5})^2 + 2(\sqrt{3})^2$
Since $(\sqrt{5})^2 = 5$ and $(\sqrt{3})^2 = 3$: $2(5) + 2(3) = 10 + 6 = 16$
Answer : The value is 16, which is a whole number, so the expression is RATIONAL.

2. Given equation: $(a^2 + 4)x^2 + 13x + 4a = 0$
For $ax^2 + bx + c = 0$, Product of zeroes = c/a
Here Product of zeroes = $4a / (a^2 + 4)$
Since the zeroes are reciprocals, their product = 1: $4a / (a^2 + 4) = 1$
Cross multiply : $4a = a^2 + 4$
Rearrange: $a^2 - 4a + 4 = 0$
This is a perfect square : $(a - 2)^2 = 0$
Answer: $a = 2$

3. Given : $2x + ky = 1$ and $5x + (k + 4)y = 2$
So $a_1=2, b_1=k, c_1=1$ and $a_2=5, b_2=(k + 4), c_2=2$
Set $a_1 / a_2 = b_1 / b_2$:
 $2/5 = k / (k + 4)$
Cross multiply : $2(k + 4) = 5k$
 $2k + 8 = 5k$
 $8 = 3k$
 $k = 8/3$
Quick check : $c_1 / c_2 = 1/2$. With $k = 8/3, a_1 / a_2 = 2/5$ which is NOT equal to $1/2$, so the 'no solution' condition is satisfied.
Answer : $k = 8/3$

4. For $3x^2 + 7x + p = 0$, Sum of roots = $-b/a = -7/3$
Let the two roots be -2 (given) and r (unknown).
 $-2 + r = -7/3$
 $r = -7/3 + 2$
 $r = -7/3 + 6/3 = -1/3$
Answer : The other root is $-1/3$

5. Let HCF = h. We are told LCM = 14h (LCM is 14 times the HCF).
 Also given: LCM + HCF = 600
 So $14h + h = 600$
 $15h = 600$
 $h = 40 \rightarrow$ so HCF = 40, and LCM = $14 \times 40 = 560$
 Now use LCM $\times$ HCF = Product of the two numbers : $560 \times 40 = 22400$
 One number is given as 280. To find the other number, divide :
 Other number = $22400 \div 280 = 80$
Answer : The other number is 80

SECTION-B

6. Simplify $\frac{1}{(2+\sqrt{3})}$ by rationalizing

$$\text{multiplying this by } (2 - \sqrt{3}) \Rightarrow \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} \Rightarrow \frac{2-\sqrt{3}}{[(2)^2-(\sqrt{3})^2]} \Rightarrow \frac{2-\sqrt{3}}{4-3} \Rightarrow \frac{2-\sqrt{3}}{1} \Rightarrow 2 - \sqrt{3}$$

Now, Let $2 - \sqrt{3}$ be a rational no. $2 - \sqrt{3} = \frac{p}{q}$ where ($q \neq 0$)

$$\sqrt{3} = 2 - \frac{p}{q}$$

$$\sqrt{3} = \frac{2q - p}{q}$$

Where, $2q - p$ & q are rational no.

And $\sqrt{3}$ is irrational

But irrational and rational no. Can not be equal

So, this contraction is due to our wrong assumption

Hence, $2 - \sqrt{3}$ i.e. $\frac{1}{(2+\sqrt{3})}$ is irrational no.

7. **Given polynomial** : $3x^2 - 7x + 2 = 0$, with zeroes α and β

Sum of zeroes ($\alpha + \beta$) = $-b/a = 7/3$

Product of zeroes ($\alpha\beta$) = $c/a = 2/3$

Let the new zeroes be : $S_1 = (2\alpha + 3\beta)$ and $S_2 = (3\alpha + 2\beta)$

- **Sum of new zeroes** ($S_1 + S_2$) :

$$(2\alpha + 3\beta) + (3\alpha + 2\beta) = 5\alpha + 5\beta = 5(\alpha + \beta)$$

$$5 \times \frac{7}{3} = \frac{35}{3}$$

- **Product of new zeroes** ($S_1 \times S_2$) :

$$(2\alpha + 3\beta)(3\alpha + 2\beta) = 6\alpha^2 + 4\alpha\beta + 9\alpha\beta + 6\beta^2 = 6(\alpha^2 + \beta^2) + 13\alpha\beta$$

Using $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$:

$$6[(\alpha + \beta)^2 - 2\alpha\beta] + 13\alpha\beta = 6(\alpha + \beta)^2 + \alpha\beta$$

$$6\left(\frac{7}{3}\right)^2 + \frac{2}{3} = 6\left(\frac{49}{9}\right) + \frac{2}{3} = \frac{98}{3} + \frac{2}{3} = \frac{100}{3}$$

The formula for a quadratic polynomial is $k[x^2 - (\text{sum of zeroes})x + (\text{product of zeroes})]$.

$$x^2 - \frac{35}{3}x + \frac{100}{3}$$

Multiplying by 3 to clear the fractions, the required polynomial is : **$3x^2 - 35x + 100$**

8. Let x be the total distance in kilometers. For both companies, the first kilometer has a fixed charge, and the remaining distance is $(x - 1)$ km.

- **Company A Fare (F_A) :** $60 + 10(x - 1) = 50 + 10x$
- **Company B Fare (F_B) :** $40 + 15(x - 1) = 25 + 15x$

1. Distance where fares are equal :

$$\text{Set } F_A = F_B :$$

$$50 + 10x = 25 + 15x$$

$$25 = 5x$$

$$x = 5 \text{ km}$$

2. When is Company B cheaper?

$$\text{Set } F_B < F_A :$$

$$25 + 15x < 50 + 10x$$

$$5x < 25$$

$$x < 5 \text{ km}$$

Company B is cheaper for distances **less than 5 km** (but at least 1 km based on the structure).

9. Equation : $3\sqrt{3}x^2 + 10x + \sqrt{3} = 0$

1. Product of terms : $(3\sqrt{3}) \times (\sqrt{3}) = 3 \times 3 = 9$

- 2. Split 10x :** We need two numbers that multiply to 9 and add to 10 (9 and 1).

$$3\sqrt{3}x^2 + 9x + x + \sqrt{3} = 0$$

$$3\sqrt{3}x(x + \sqrt{3}) + 1(x + \sqrt{3}) = 0$$

$$(3\sqrt{3}x + 1)(x + \sqrt{3}) = 0$$

$$\text{Roots : } x = -\frac{1}{3\sqrt{3}} \text{ (or } -\frac{\sqrt{3}}{9}) \text{ and } x = -\sqrt{3}.$$

10. Given : $2x^2 - 5x + k$ with zeroes α and β .

- **Sum of zeroes :** $\alpha + \beta = -(-5)/2 = 5/2$

- **Product of zeroes :** $\alpha\beta = k/2$

Equation provided : $\alpha^2 + \beta^2 + \alpha\beta = 21/4$

Rewrite $\alpha^2 + \beta^2$ as $(\alpha + \beta)^2 - 2\alpha\beta$

$$(\alpha + \beta)^2 - 2\alpha\beta + \alpha\beta = \frac{21}{4}$$

$$(\alpha + \beta)^2 - \alpha\beta = \frac{21}{4}$$

Substitute values :

$$\left(\frac{5}{2}\right)^2 - \frac{k}{2} = \frac{21}{4}$$

$$\frac{25}{4} - \frac{k}{2} = \frac{21}{4}$$

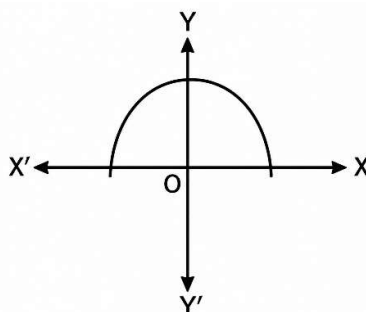
$$\frac{25-21}{4} = \frac{k}{2}$$

$$1 = \frac{k}{2}$$

$$k = 2$$

SECTION-C

11. i. We have, $f(x) = ax^2 + bx + c$, $a < 0$ It means a figure is a shape of parabola which open downwards.



- ii. Since, $x = 4$ is one of the zeroes of the polynomial $(p - 1)x^2 + px + 1$.
 $\therefore (p - 1)(4)^2 + p(4) + 1 = 0$
 $\Rightarrow 16p - 16 + 4p + 1 = 0$
 $\Rightarrow 20p = 15$
 $\Rightarrow p = 3/4$
- iii. Required quadratic polynomial = $k[x^2 - (\text{Sum of zeroes})x + (\text{Product of zeroes})] = k(x^2 - (-12 + 5)x + (-12)(5)) = k(x^2 + 7x - 60)$, where k is any arbitrary constant.
- iv. Let the zeroes of the quadratic polynomial be a . Since, one zero of the polynomial $f(x)$ is reciprocal of the other
 $\therefore$ Product of zeroes = $(-1)^2 \times (\text{Constant term/Coefficient of } x^2)$
 $\Rightarrow a \times (1/a) = 1 \times (m/5) \Rightarrow m = 5$
12. (i) Let's actually calculate $\alpha + \beta$ and $\alpha\beta$ ourselves and compare with Pooja's numbers.

$$\alpha = \frac{3}{\sqrt{2}} \text{ and } \beta = -\sqrt{2}/5$$

Sum: $3/\sqrt{2} + (-\sqrt{2}/5)$. To add these, take LCM of denominators ($\sqrt{2}$ and 5), which is $5\sqrt{2}$:

$$\frac{3}{\sqrt{2}} = \frac{15}{(5\sqrt{2})} \text{ and } \frac{\sqrt{2}}{5} = \frac{2}{(5\sqrt{2})} [\text{because } \sqrt{2} \times \sqrt{2} = 2]$$

$$\text{So Sum} = \frac{15}{(5\sqrt{2})} - \frac{2}{(5\sqrt{2})} = \frac{13}{(5\sqrt{2})} - \text{this matches what Pooja got!}$$

$$\text{Product : } (3\sqrt{2}) \times (-\sqrt{2}/5) = -(3 \times \sqrt{2})/(\sqrt{2} \times 5) = -\frac{3}{5} \text{ this also matches Pooja's value!}$$

So Neha is WRONG - Pooja's calculations of Sum and Product were correct, and so was her final polynomial.

- (ii) Substitute $x = \frac{3}{\sqrt{2}}$ into $5\sqrt{2}x^2 - 13x - 3\sqrt{2}$ and see if we get 0 :

$$5\sqrt{2} \times (3/\sqrt{2})^2 - 13 \times (3/\sqrt{2}) - 3\sqrt{2}$$

$$= 5\sqrt{2} \times 9/2 - 39/\sqrt{2} - 3\sqrt{2} = 45\sqrt{2}/2 - 39\sqrt{2}/2 - 6\sqrt{2}/2 \text{ (converting all terms to have denominator 2 with } \sqrt{2})$$

$$= (45 - 39 - 6) \frac{\sqrt{2}}{2} = 0$$

$$\text{Now substitute } x = -\frac{\sqrt{2}}{5}$$

$$5\sqrt{2} \times (2/25) - 13 \times (-\sqrt{2}/5) - 3\sqrt{2} = 10\sqrt{2}/25 + 13\sqrt{2}/5 - 3\sqrt{2}$$

$$= 2\sqrt{2}/5 + 13\sqrt{2}/5 - 15\sqrt{2}/5 = 0$$

Since both zeroes satisfy the polynomial, Pooja's polynomial $5\sqrt{2}x^2 - 13x - 3\sqrt{2} = 0$ is confirmed correct.

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (13/(5\sqrt{2}))^2 - 2(-3/5)$

$$(13/(5\sqrt{2}))^2 = \frac{169}{(25 \times 2)} = \frac{169}{50} - 2 \times (-3/5) = +6/5 = 60/50$$

$$\alpha^2 + \beta^2 = 169/50 + 60/50 = 229/50$$

(iv) The Factor Theorem says : if $(x - a)$ is a factor of a polynomial $q(x)$, then $q(a) = 0$.

Here $(x+3)$ is a factor, which is the same as $(x - (-3))$, so $a = -3$.

Therefore $q(-3) = 0$

13. Let the original fraction be x/y (x = numerator, y = denominator).

(i) '2 added to numerator, 2 subtracted from denominator, fraction becomes 2' : $(x+2)/(y-2) = 2$

Cross multiply: $x + 2 = 2(y-2) = 2y - 4$

Equation 1: $x - 2y + 6 = 0$

(ii) '2 added to denominator only, fraction becomes 1/3' : $x/(y+2) = 1/3$

Cross multiply: $3x = y + 2$

Equation 2: $3x - y - 2 = 0$

(iii) From Equation 2: $y = 3x - 2$

Substitute this into Equation 1 ($x - 2y + 6 = 0$): $x - 2(3x-2) + 6 = 0$

$x - 6x + 4 + 6 = 0 - 5x + 10 = 0 \quad x = 2$

Put $x=2$ back into $y = 3x-2$: $y = 3(2)-2 = 4$

Original fraction = $x/y = 2/4 = 1/2$

Check : $(2+2)/(4-2) = 4/2 = 2 \checkmark$ and $2/(4+2) = 2/6 = 1/3 \checkmark$ — both conditions match!

(iv) For no solution : $a_1/a_2 = b_1/b_2 \neq c_1/c_2$

Given: $4x + y = 2$ and $(3k-1)x + (k-3)y = 3k+5$

$a_1 = 4, b_1 = 1, a_2 = (3k-1), b_2 = (k-3) \quad 4/(3k-1) = 1/(k-3)$

Cross multiply : $4(k-3) = 1 \times (3k-1) \quad 4k - 12 = 3k - 1$

$4k - 3k = -1 + 12$

$k = 11$

14. Let the actual (uniform) speed = x km/h and the scheduled time = t hours.

So the distance = $x \times t$ (this stays the same in every case, since it's the same journey).

Case 1: 'If speed had been 6 km/h faster, it takes 4 hours LESS':

New speed = $(x+6)$, New time = $(t-4)$, and distance is still the same: $(x+6)(t-4) = xt$

Expand the left side: $xt - 4x + 6t - 24 = xt$

The xt on both sides cancels out:

$-4x + 6t - 24 = 0 \rightarrow 6t - 4x = 24 \rightarrow$ dividing by 2: $3t - 2x = 12$...Equation (i)

Case 2: 'If speed had been 6 km/h slower, it takes 6 hours MORE':

New speed = $(x-6)$, New time = $(t+6)$:

$$(x-6)(t+6) = xt$$

Expand: $xt + 6x - 6t - 36 = xt$

Again xt cancels:

$$6x - 6t - 36 = 0 \rightarrow 6x - 6t = 36 \rightarrow \text{dividing by 6: } x - t = 6 \rightarrow t = x - 6 \text{ ...Equation (ii)}$$

Now substitute Equation (ii) into Equation (i):

$$3(x-6) - 2x = 12$$

$$3x - 18 - 2x = 12$$

$$x - 18 = 12$$

$$x = 30 \text{ km/h}$$

Now find t using Equation (ii): $t = 30 - 6 = 24$ hours

Finally, Distance = Speed $\times$ Time = $x \times t = 30 \times 24$

Answer: Length of the journey = 720 km

15. (i) For $5x^2 - 4x - 3 = 0$: here $a=5$, $b=-4$, $c=-3$

$$D = (-4)^2 - 4(5)(-3) = 16 + 60 = 76$$

Since $76 > 0$:

Answer: Roots are real and distinct

- (ii) For $x^2 + 3x + 3 = 0$: here $a=1$, $b=3$, $c=3$

$$D = (3)^2 - 4(1)(3) = 9 - 12 = -3$$

Since $-3 < 0$:

Answer: Roots are not real (no real roots exist)

- (iii) For $3x^2 - 2x + 4k = 0$ to have equal roots, we need $D = 0$

Here $a=3$, $b=-2$, $c=4k$

$$D = (-2)^2 - 4(3)(4k) = 4 - 48k$$

$$\text{Set } D = 0: 4 - 48k = 0$$

$$48k = 4$$

$$k = 4/48 = 1/12$$

- (iv) For $x^2 + 5kx + 16 = 0$ to have NO real roots, we need $D < 0$

Here $a=1$, $b=5k$, $c=16$

$$D = (5k)^2 - 4(1)(16) = 25k^2 - 64$$

$$\text{Set } D < 0: 25k^2 - 64 < 0$$

$$25k^2 < 64$$

$$k^2 < 64/25$$

Taking square root on both sides (this gives a range, not just one value):

$$-8/5 < k < 8/5$$

